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PROJECT CHARTER & PROBLEM STATEMENT

AayurLink

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PROJECT METADATA

Project Name : AayurLink
Domain : Digital Health / Chronic Kidney Disease (CKD/CKDu) Care
Project Role : Lead Business Analyst & Project Manager
Document Version : 1.0
Status : Approved / Active Baseline

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1. EXECUTIVE SUMMARY

AayurLink is a centralized, mobile-responsive digital healthcare platform engineered to streamline hemodialysis access, track longitudinal renal biomarkers, and provide visibility into verified community drinking water sources for Chronic Kidney Disease (CKD/CKDu) patients across Sri Lanka. The system connects patients, rural caregivers, and nephrology clinic teams into an integrated, real-time workflow.

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2. BUSINESS PROBLEM STATEMENT

2.1 Background:
Endemic regions in Sri Lanka—such as the North Central, Uva, and North Western provinces—experience high concentrations of CKDu. Most patients are rural, low-income agrarian workers dependent on state hospitals for life-sustaining dialysis and clinical supervision.

2.2 Core Operational Bottlenecks:
1. Dialysis Scheduling & Queuing Chaos:
   - Due to limited machine capacity, patients arrive as early as 3:00 AM and endure unreserved waiting times between 6 to 8 hours.
   - Manual paper-based ticketing results in queue disputes, extreme physical fatigue for fragile patients, and high recurring transport expenses.
2. Fragmented & Lost Diagnostic Records:
   - Serum Creatinine, eGFR, and electrolyte lab reports are maintained on loose paper slips, resulting in frequent misplacement or degradation.
   - Clinicians cannot rapidly evaluate historical disease progression trends.
3. Information Gap in Clean Drinking Water Access:
   - Patients lack verified, real-time information regarding the functional and maintenance status of nearby community Reverse Osmosis (RO) water stations.

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3. PROJECT OBJECTIVES & SUCCESS METRICS (SMART KPIS)

- OBJ-01: Reduce average hospital waiting times from 5.5 hours to under 1.5 hours (>60% reduction via scheduled time slots).
- OBJ-02: Achieve 100% digital logging and automated line-graph visualization for patient Creatinine and eGFR trends.
- OBJ-03: Ensure rural accessibility by enabling task completion (slot booking) in Sinhala within 3 steps / under 45 seconds.
- OBJ-04: Reduce clinical nursing administrative overhead for daily patient check-ins by 50%.

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4. SCOPE BASELINE

4.1 In-Scope (Phase 1: MVP Delivery):
- Authentication & Profiles: Phone-based login with role access (Patient/Caregiver vs. Hospital Nurse/Admin) and bilingual support (Sinhala & English).
- Dialysis Slot Booking: Morning and Afternoon shift slot allocation, booking confirmation, and visual availability display.
- Vital Health Tracker: Input interface for Serum Creatinine and eGFR with an automated chronological trajectory line graph.
- RO Plant Directory: District-filtered listing and status overview of operational community clean drinking water plants.
- Hospital Clinic Portal: Daily patient roster view and single-tap digital check-in.

4.2 Out-of-Scope (Deferred to Future Releases):
- Real-time video telemedicine consultations.
- E-pharmacy integration and home medication courier logistics.
- Organ transplant recipient-donor legal matching database.
- Integration with external legacy hospital software (e.g., HHIMS).

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5. STAKEHOLDER MATRIX

- Primary Users : CKD/CKDu Patients and Family Caregivers (Booking, viewing logs).
- Admin Users : Hemodialysis Unit Nurses (Bed/slot management, check-in validation).
- Clinical Experts : Consultant Nephrologists (Biomarker threshold standards).
- Execution Lead : Lead Business Analyst & Project Manager (Requirements, delivery).

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6. CONSTRAINTS & ASSUMPTIONS

- Assumptions: Users or family members have access to a basic smartphone with a browser; clinics possess at least one web-enabled terminal.
- Constraints: 4-week execution lifecycle (4–5 hours daily dedication); zero-cost budget utilizing free-tier hosting and open-source stacks.